Project Report

On

Design of a 6-Bit Analog-to-Digital Converter

Course No: ECE 2104

Course Name: Digital Electronics and Logic Circuits Laboratory

Group: A-10

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Problem Statement

Design and implement a 6-bit Analog-to-Digital Converter (ADC) with a Full-Scale Range (FSR) not exceeding 8V:

Open-ended features:

1. Select standard, commercially available values of resistors and capacitors for the circuit implementation.
2. Any suitable circuit configuration or conversion technique may be used to implement the ADC.
3. Any conventional circuit simulation software may be used to simulate and verify the designed circuit.

Result analysis:

1. Observe the output signals at each stage of the circuit and provide appropriate explanations of their functions. Determine the voltage resolution of the ADC.
2. Measure the output voltage and digital output codes for different analog input voltages and compare them with the theoretically calculated values.
3. Compare the simulated results with the calculated results.
4. Verify whether the simulated and experimental results agree with the theoretical calculations and explain any discrepancies.
5. Evaluate the performance of the designed ADC in terms of conversion accuracy, output stability, and its ability to convert analog input voltages into the corresponding 6-bit digital representation.

Evaluation

	Excellent	Above Average	Average	Below Average	Marks (10)
Organization/Presentation					
Necessary data presentation					
Result Analysis					
Conclusion					

Signature of Examiner

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1. Objectives

1. To design and implement a 6-bit Analog-to-Digital Converter (ADC) with a full-scale input range not exceeding 8 V.
2. To understand the working principle of analog-to-digital conversion and the relationship between analog input voltage and digital output code.
3. To calculate the theoretical voltage resolution of the designed ADC.
4. To observe and analyze the output signals at different stages of the circuit.
5. To compare the theoretical and simulated results and evaluate the performance of the designed ADC.

2. Introduction

An Analog-to-Digital Converter (ADC) is an electronic circuit that converts a continuous analog signal into a digital representation. It serves as an interface between the analog and digital domains, allowing digital systems such as computers, microcontrollers, and digital instruments to process real-world signals. ADCs are widely used in data acquisition systems, industrial instrumentation, communication systems, and electronic measurement devices.

The operation of an ADC involves quantization and encoding. During quantization, the analog input voltage range is divided into discrete levels, and each level is assigned a corresponding binary code. The resolution of an ADC depends on the number of output bits, which determines how finely the input voltage range can be represented. A 6-bit ADC provides 64 distinct quantization levels, producing digital output codes ranging from 000000 to 111111.

In this project, a 6-bit ADC with a full-scale input range not exceeding 8 V is designed and implemented. The circuit is simulated to investigate its working principle and observe the output signals at different stages. The relationship between the analog input voltage and the corresponding digital output is examined to evaluate the conversion process.

Furthermore, the theoretical characteristics of the designed ADC are compared with the simulation and practical results. This comparison helps verify the circuit's functionality, assess its performance, and identify possible discrepancies caused by component tolerances, circuit limitations, and non-ideal operating characteristics.

3. Theory

An Analog-to-Digital Converter (ADC) is an electronic circuit that converts an analog input voltage into a corresponding digital output. In this project, a 6-bit Counter-Type ADC is designed for an input range of 0–8 V. The circuit uses two cascaded 74HC161 counters, an R–2R ladder

network, an operational amplifier, and a comparator to perform the conversion. Each section contributes to generating the reference voltage, comparing it with the input, and producing the final 6-bit digital output.

3.1 Counter Section

A digital counter is a sequential logic circuit that advances its binary output in response to clock pulses. The 74HC161 is a 4-bit synchronous binary counter with 16 possible states, ranging from 0000 to 1111.

In this project, two 74HC161 counters are cascaded to obtain a 6-bit output. The counters are reset to zero at the beginning of a conversion and increment with each clock pulse. The six selected output bits represent the digital conversion result, while the comparator controls the counting process.

The number of possible digital output codes is

$$N = 2^n$$

For a 6-bit ADC,

$$N = 2^6 = 64$$

Thus, the output ranges from decimal 0 to 63. The approximate conversion time is determined by the number of counting steps and the clock frequency:

$$t_{conv} \approx \frac{D}{f_{clk}}$$

where *D* is the number of counting steps and f_{clk} is the clock frequency.

3.2 R–2R Ladder Network

An R–2R ladder is a resistor network that uses only two resistor values, R and 2R, to convert a digital binary input into a proportional analog voltage. Each digital bit contributes a weighted voltage according to its binary significance.

For an ideal 6-bit DAC using an R–2R ladder, the output voltage can be expressed as

$$V_{DAC} = V_{ref} \times \left(\frac{b_5}{2} + \frac{b_4}{4} + \frac{b_3}{8} + \frac{b_2}{16} + \frac{b_1}{32} + \frac{b_0}{64}\right)$$

where b_5 is the most significant bit, b_0 is the least significant bit, and each bit is either 0 or 1.

For a conventional unipolar DAC, the equivalent expression is

$$V_{DAC} = V_{ref} \times \frac{D}{2^n}$$

where D is the decimal input code. The ideal voltage step is

$$\Delta V = \frac{V_{ref}}{2^n}$$

The R–2R ladder provides a simple and systematic method of generating an analog voltage from the counter output. Its accuracy depends on resistor matching, loading, and the characteristics of the connected amplifier.

3.3 Operational Amplifier Section

An operational amplifier may be used to scale or buffer the voltage generated by the R–2R ladder. In the designed circuit, the amplifier is configured as a non-inverting amplifier with a feedback network to increase the ladder output voltage.

The theoretical voltage gain is

$$A_v = 1 + \left(\frac{R_f}{R_g}\right)$$

where R_f is the feedback resistance and R_g is the resistance connected between the inverting input and ground.

The output voltage is given by

$$V_{out} = A_v \times V_{in}$$

For a voltage gain of approximately 2,

$$V_{out} = 2 \times V_{in}$$

Therefore, a 4 V input to the amplifier ideally produces an 8 V output. In practice, the output swing is limited by the operational amplifier's supply voltage, load, and internal characteristics. The amplifier must remain within its operating limits to provide a suitable voltage for comparison.

3.4 Comparator Section

A comparator compares two input voltages and changes its output state according to their relative magnitudes. In this project, it compares the conditioned analog input voltage with the voltage generated by the R–2R ladder and amplifier stages.

The comparison condition for stopping the count is

$$V_{DAC,out} \geq V_{in}$$

where $V_{DAC,out}$ is the conditioned ladder output and V_{in} is the analog input voltage. The actual stopping polarity depends on the comparator connections and the logic used to control the counters.

When the threshold is reached, the comparator changes state and disables further counting. The final counter output then represents the digital code corresponding to the analog input. Comparator offset voltage, propagation delay, and output-interface characteristics may affect the conversion threshold and final result.

3.5 ADC Resolution and Digital Output Calculation

The resolution of an ADC indicates the nominal voltage increment represented by one least significant bit (LSB). It is calculated using

$$\Delta V = \frac{V_{FSR}}{2^n}$$

For the 6-bit ADC with a full-scale input range of 8 V,

$$\Delta V = \frac{8}{2^6} = \frac{8}{64} = 0.125\ V$$

Thus, the nominal resolution is 125 mV per LSB.

The theoretical decimal output code for an ideal unipolar ADC can be calculated as

$$D = \left(\frac{V_{in}}{\Delta V}\right)$$

with the result limited to the range 0–63. For example, when the input voltage is 3 V,

$$D = \left(\frac{3}{0.125}\right) = 24$$

The corresponding binary output is 011000. The highest output code, 111111, represents decimal 63 and corresponds nominally to 7.875 V under the ideal quantization convention.

The counter, R–2R ladder, operational amplifier, and comparator work together to perform analog-to-digital conversion. The counter generates successive digital codes, the ladder converts these codes into corresponding analog voltage levels, the amplifier conditions the voltage, and the comparator determines when the conversion threshold has been reached. The theoretical relationships described above provide the basis for calculating the ADC resolution,

predicting digital output codes, and evaluating the simulated and practical performance of the designed circuit.

4. Apparatus

Table-01: Apparatus Required for 6-Bit Analog-to-Digital Converter

Sl. No.	Name of Apparatus	Rating	Quantity
01	DC Power Supply	4V, 9V, (0-8)V	03
02	Function Generator	(1-10)kHz, $2V_{p-p}$(max)	01
03	4-bit Counter IC (74HC161)	(2-6)V, 500mA(max)	02
04	AND Gate (7408)	5.5V(max), 800mA(max)	01
05	Resistor	10kΩx30, 330Ωx6	36
06	Op-Amp IC	LM358, IL393	02
07	Breadboard	-----	02
08	LEDs	3-3.6V, 20mA(max)	06
09	Push Button	5V(max), 1mA	01
10	Connecting Wires	----	as required

5. Circuit Diagram

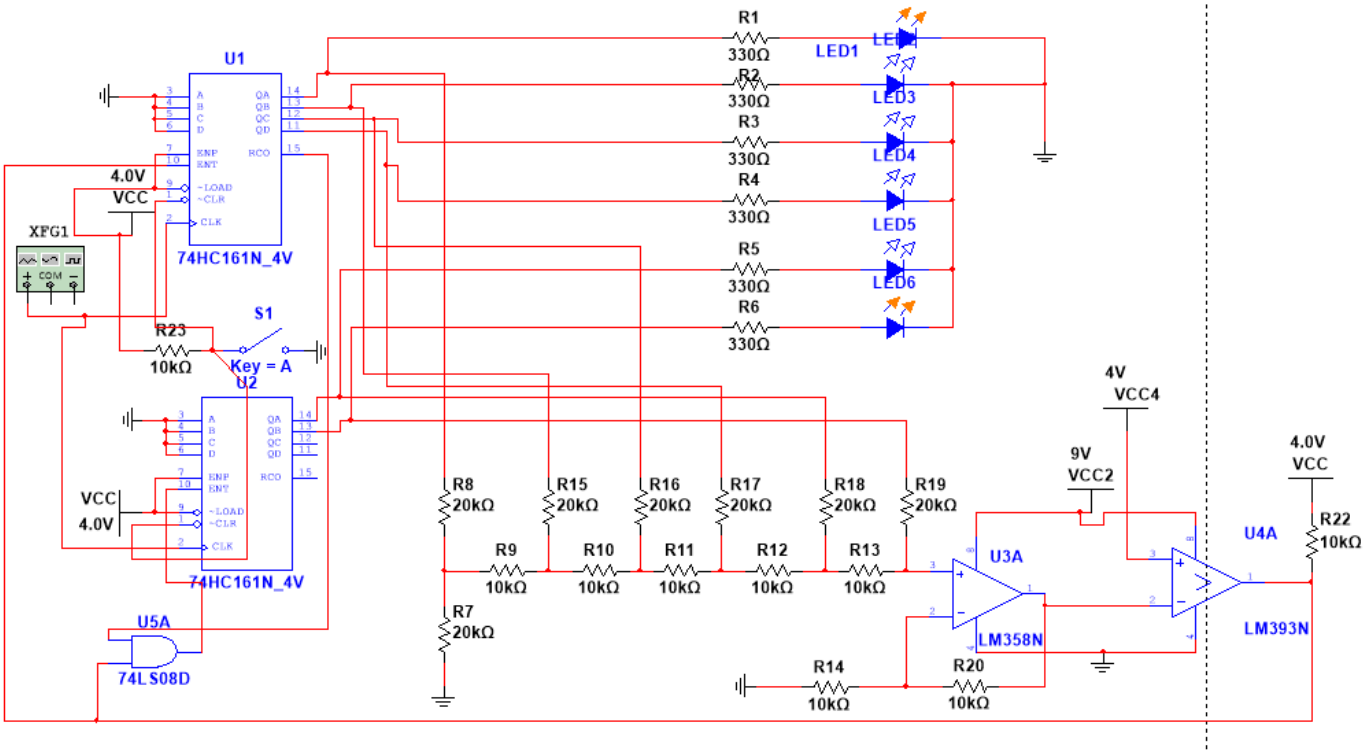


Fig-01: Simulated Circuit Diagram for the ADC (Multisim)

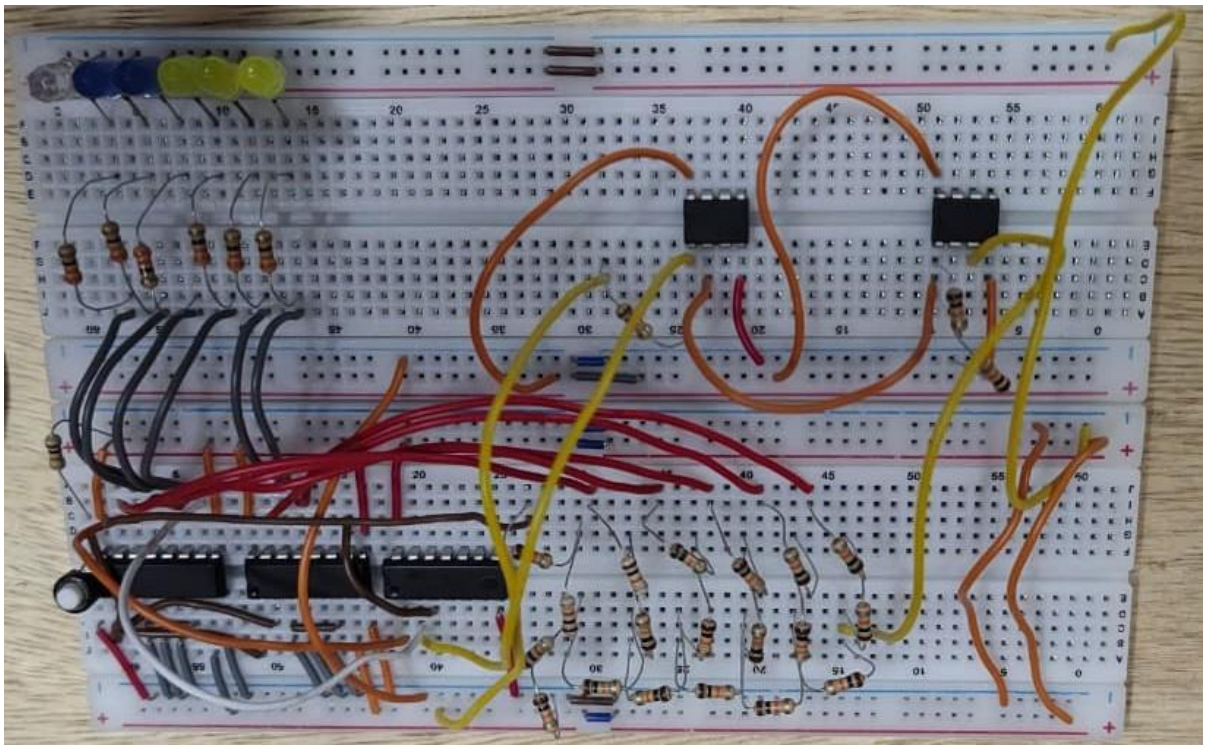


Fig-02: Real-World Circuit Diagram for the ADC

6. Experimental Data

Table-02: Circuit Measurements for the Counter-Type ADC

Input Voltage	Measured Binary Sequence	Measured Output Decimal	Theoretical Output Decimal	Measured Output Voltage
1.16 V	001010	9	9.28	1.21 V
2 V	010001	17	16	2.09 V
3.01 V	011001	25	24.08	3.05 V
4.02 V	100010	34	32.16	4.16 V
5.06 V	101010	42	40.48	5.13 V
5.35 V	101101	45	42.8	5.43 V
6.1 V	110011	51	49	6.24 V
6.38 V	110110	54	51.04	6.53 V
7.31 V	111101	61	58.48	7.44 V
7.6 V	111111	63	61	7.67 V

Table-03: Simulated Circuit Measurements for the Counter-Type ADC

Input Voltage	Digital Output	Decimal Output	DAC Voltage	Amplifier Output Voltage
1.16 V	001010	9	625 mV	1.25 V
2 V	010001	17	1.06 V	2.13 V
3.01 V	011001	25	1.56 V	3.13 V
4.02 V	100001	33	2.06V	4.13 V
5.06 V	101001	41	2.56 V	5.13 V
5.35 V	101011	43	2.69 V	5.38 V
6.1 V	110001	49	3.06 V	6.13 V
6.38 V	110100	52	3.25 V	6.5 V
7.31 V	111011	59	3.69 V	7.38 V
7.6 V	111101	61	3.81 V	7.62 V

Table-04: Error Calculations for the Counter-Type ADC

Input (V)	*Ideal Code*	*Measured Code*	*Code Error*	*Error (%)*
1.16	9	9	0	0.00%
2.00	16	17	+1	6.25%
3.01	24	25	+1	4.17%
4.02	32	34	+2	6.25%
5.06	40	42	+2	5.00%
5.35	42	45	+3	7.14%
6.12	48	51	+3	6.25%
6.38	51	54	+3	5.88%
7.31	58	61	+3	5.17%
7.60	60	63	+3	5.00%

7. Calculations

Given,

- Number of bits, $n = 6$
- Full-scale input range, $V_{FS} = 8V$
- Reference/resistor value, $R = 10\,k\Omega$
- $2R = 20\,k\Omega$
- Amplifier feedback resistor, $R_f = 10\,k\Omega$
- Amplifier grounding resistor, $R_g = 10\,k\Omega$
- Clock frequency, $f = 1\,Hz$

7.1 Number of Quantization Levels and ADC Resolution

Number of levels,

$$N = 2^n = 2^6 = 64$$

Resolution,

$$\Delta V = \frac{V_{FS}}{N}$$

$$\Delta V = \frac{8}{64} = 0.125\,V$$

Therefore, **ADC resolution = 125 mV/LSB.**

7.2 Theoretical Digital Output and R–2R Ladder Calculation

Theoretical digital code,

$$D = \frac{V_{in}}{\Delta V}$$

For $V_{in} = 3.01\,V$,

$$D = \frac{3.01}{0.125} = 24.08$$

Taking the integer part,

$$D = 24 = (011000)_2$$

Therefore, the theoretical digital output is 011000.

Given,

$$R = 10\,k\Omega, \qquad 2R = 20\,k\Omega$$

Resistor ratio,

$$\frac{2R}{R} = \frac{20\,k\Omega}{10\,k\Omega} = 2$$

For an ideal 6-bit DAC output,

$$V_{DAC} = V_{REF} \times \frac{D}{64}$$

The actual output depends on the reference voltage and the ladder connection.

7.3 Amplifier Gain

For a non-inverting amplifier,

$$A_v = 1 + \frac{R_f}{R_g}$$

Substituting the resistor values,

$$A_v = 1 + \frac{10\,k\Omega}{10\,k\Omega} = 2$$

Therefore, the designed voltage gain is 2.

7.4 Net Error Calculation

Net code error,

$$E_{net} = \sum D_{measured} - \sum D_{ideal}$$

Using the listed decimal codes from the experimental data, the net code error is obtained by summing the individual code errors.

$$E_{net} = 0 + 1 + 1 + 2 + 2 + 3 + 3 + 3 + 3 + 3 = 21$$

Therefore, net code error = +21.

Net percentage error,

$$E_{net}(\%) = \frac{21}{389} \times 100 = 5.40\%$$

Therefore, the net percentage error is 5.40%.

8. Result Analysis

8.1 Theoretical, Simulated, and Practical Digital Output Comparison

The performance of the designed 6-bit counter-type ADC was evaluated by comparing theoretical digital outputs with simulated and practical experimental results. The ADC has an input range of 0–8 V and 64 quantization levels, giving a theoretical resolution of 0.125 V/LSB. The ideal digital code was calculated using:

Ideal Code = Integer part of $\left(\frac{Input\ Voltage}{0.125\ V}\right)$

Table-05: Comparison of Theoretical, Simulation and Practical Decimal Equivalent Code

Input (V)	Ideal Code	Simulated Code	Practical Code
1.16	9	9	9
2.00	16	17	17
3.01	24	25	25
4.02	32	33	34
5.06	40	41	42
5.35	42	43	45
6.10	48	49	51
6.38	51	52	54
7.31	58	59	61
7.60	60	61	63

The comparison indicates that the simulated outputs are generally close to the ideal values, while the practical outputs show larger positive deviations at higher input voltages. These differences may result from analog conversion errors, comparator switching thresholds, component tolerances, or measurement uncertainty.

8.2 Analysis of ADC Resolution and Quantization

The theoretical resolution is:

Resolution = Full-scale input range / Number of levels

Resolution = $\frac{8\,V}{64}$ = 0.125 V/LSB

Thus, one increment in the ideal digital output represents an input change of approximately 125 mV. The theoretical output ranges from code 0 (000000) to code 63 (111111), with code 63 representing 7.875 V under ideal scaling.

The experimental results show increasing differences between the ideal and practical codes as the input voltage increases. At 7.60 V, the ideal code is 60, whereas the practical table records code 63. This corresponds to a difference of three codes, equivalent to 375 mV under ideal scaling.

8.3 Analysis of Simulated and Practical Error

Code Error = Measured Code − Ideal Code

Percentage Error = $\left(\frac{|Code\ Error|}{Ideal}\right) \times 100\%$

Table-06: Error Calculation for Simulated and Practical Inputs

Input (V)	Simulated Error	Simulated Error (%)	Practical Error	Practical Error (%)
1.16	0	0.00%	0	0.00%
2.00	+1	6.25%	+1	6.25%
3.01	+1	4.17%	+1	4.17%
4.02	+1	3.13%	+2	6.25%
5.06	+1	2.50%	+2	5.00%
5.35	+1	2.38%	+3	7.14%
6.10	+1	2.08%	+3	6.25%
6.38	+1	1.96%	+3	5.88%
7.31	+1	1.72%	+3	5.17%
7.60	+1	1.67%	+3	5.00%

The simulated error is zero or one code based on the recorded decimal values. The practical error increases to three codes at higher input levels. Although the percentage error generally decreases as the ideal code increases, the absolute code deviation remains important when evaluating the converter.

The total signed code error across the ten test points is:

- Simulated net error = +9 codes
- Practical net error = +21 codes

Using the sum of the ideal codes, 389, as the denominator:

Simulated net error (%) = $\left(\frac{9}{389}\right) \times 100$ = 2.31%

Practical net error (%) = $\left(\frac{21}{389}\right)$ × 100 = 5.40%

These values represent aggregate signed code-error percentages for the selected test points, not standard full-scale accuracy specifications.

8.4 Sources of Error and Limitations

The observed differences between ideal, simulated, and practical results may be associated with the following factors:

1. Resistor tolerances: Variations in the R–2R ladder resistors can alter the DAC output voltage and effective conversion thresholds.
2. Comparator characteristics: Input offset voltage, switching behavior, and output-stage characteristics may affect the point at which the counter stops.
3. Amplifier limitations: The LM358 has non-ideal output swing and offset characteristics, which may influence analog signal levels.
4. Power-supply variation: Changes in supply voltage can affect the analog stages and logic operation.
5. Wiring and contact resistance: Breadboard connections and loose contacts can introduce additional errors in the practical circuit.
6. Measurement uncertainty: Meter accuracy, probe connections, and measurement timing can influence recorded values.
7. Data-recording inconsistencies: Errors in recording digital codes can affect the calculated error percentages and simulation-versus-practical comparison.

These are possible explanations; the available measurements do not establish which factor is responsible for the observed deviations.

Overall, the experiment demonstrated the operation and performance of the counter-type ADC. The observed differences between theoretical, simulated, and practical results highlight the influence of real circuit characteristics and measurement uncertainty on conversion accuracy.

9. Discussion

The 6-bit counter-type ADC was designed, simulated, and tested experimentally to convert analog input voltages into corresponding digital outputs. With an input range of 0–8 V, the theoretical resolution was calculated as 125 mV/LSB. The simulation results showed relatively small deviations from the ideal digital codes, while the practical measurements exhibited larger positive deviations, particularly at higher input voltages.

The R–2R ladder generated the analog voltage corresponding to the counter output, and the LM358 amplifier provided an approximately twofold voltage gain in simulation. Differences between simulated and practical results may be attributed to resistor tolerances, comparator characteristics, supply variations, wiring imperfections, and measurement uncertainty.

Overall, the experiment demonstrated the operating principle of a counter-type ADC and highlighted the differences between theoretical predictions, simulation results, and practical circuit performance.

10. Conclusion

The 6-bit counter-type ADC was successfully designed, simulated, and experimentally evaluated for an input range of 0–8 V. The theoretical resolution was found to be 125 mV/LSB, and the circuit demonstrated the conversion of analog input voltages into corresponding digital outputs. The simulation results showed better agreement with the theoretical values than the practical measurements. The observed deviations highlighted the effects of component tolerances, comparator behavior, and measurement uncertainty on conversion accuracy. Overall, the project provided practical insight into ADC operation, digital code generation, and the comparison of theoretical, simulated, and experimental results.

11. References

1. Sedra, A. S., and Smith, K. C., Microelectronic Circuits, Oxford University Press.
2. Texas Instruments, LM358 Dual Operational Amplifiers Datasheet.
3. Texas Instruments, SN74HC161A Synchronous 4-Bit Binary Counter Datasheet.
4. Texas Instruments, IL393 Dual Differential Comparators Datasheet.
5. National Instruments, Multisim User Guide.
6. ECE-2104 course materials and laboratory instructions, Khulna University of Engineering & Technology (KUET).